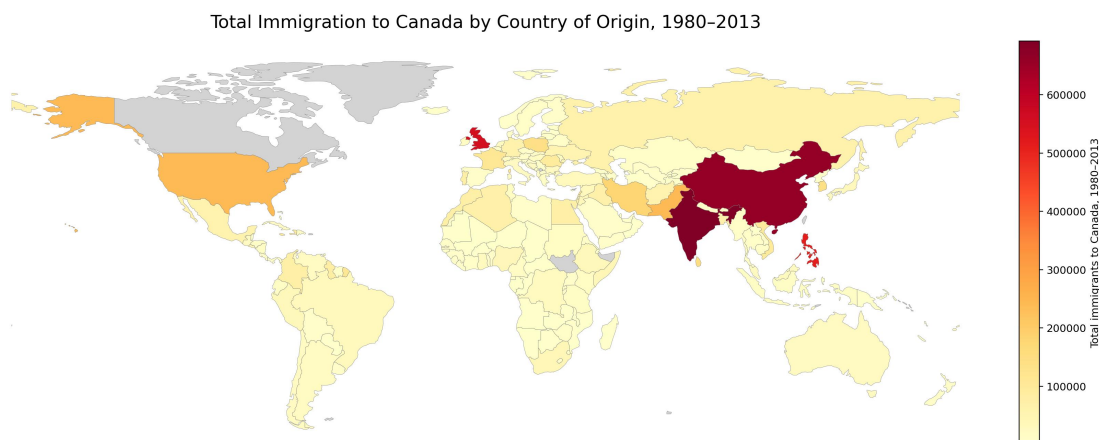


CANADA IMMIGRATION VISUAL ANALYTICS

A Python-Based Exploratory, Statistical, and Geospatial Visualization Study

Portfolio Report



Cover figure. Cumulative immigration origins to Canada, 1980–2013.

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Executive Summary

This portfolio project investigates **international immigration flows to Canada** through a deliberately visualization-centered analytical workflow. The purpose is to demonstrate how rigorous data preparation, exploratory graphics, statistical visualization, and geospatial mapping can be combined to discover structure and communicate evidence.

The analysis uses the **United Nations Population Division's International Migration Flows to and from Selected Countries: The 2015 Revision**, focusing on **annual immigration to Canada by citizenship/origin from 1980 through 2013**. After explicit scope filtering, the analytical dataset contains 194 identified international origins, 34 annual observations per origin, geographic classifications, and a cumulative immigration measure. Across these origins, the cleaned data contain 6,409,133 recorded immigrant arrivals over the full period.

The visual evidence identifies three dominant empirical patterns. **First, identified international immigration increased substantially over the long run**, from 99,137 in 1980 to 257,535 in 2013, **but the path included clear peaks, declines, and recoveries**; the annual maximum was 276,956 in 2010. **Second, immigration was concentrated among a relatively small number of origins: India, China, the United Kingdom, and the Philippines** were the four largest cumulative sources, and only eight origins were required to exceed 50% of cumulative immigration. **Third, the geographic composition shifted materially toward Asia**. Asia's annual share increased from 31.3% in 1980 to 60.2% in 2013, while Europe's share declined from 40.1% to 11.1%.

This report includes analytically distinct visualization from the related notebooks. The notebooks remain the computational record; the report organizes their figures around empirical questions, explains what each representation contributes, and synthesizes the findings.

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1. Motivation and Analytical Positioning

Exploratory data analysis is most valuable when visualization is treated as a method for interrogating data rather than as a final decorative step. The project was therefore performed a **question-driven analysis**. Each visualization contributes a distinct analytical perspective: trend, ranking, composition, distribution, concentration, variability, normalization, association, or geography.

This approach is relevant to business analytics because many applied decisions require analysts to move between exploration and explanation. A time series can reveal changing demand, a ranked bar chart can expose market concentration, a distribution plot can reveal heterogeneity, a heatmap can expose segment structure, and a geographic map can identify spatial clusters. The same reasoning used here for immigration data transfers directly to customer, market, operations, risk, and portfolio analytics.

2. Data Source, Scope, and Preparation

2.1 Source metadata

The raw file, **Canada.xlsx**, is derived from the **United Nations Department of Economic and Social Affairs, Population Division, International Migration Flows to and from Selected Countries: The 2015 Revision** (POP/DB/MIG/Flow/Rev.2015). The project uses the **Canada by Citizenship worksheet** and annual immigration-flow observations from 1980 through 2013. The analytical unit is an identified foreign citizenship/origin category observed annually; the Country field should therefore be interpreted as the source's reported citizenship/origin classification, not as immigrant stock, ethnicity, or place of birth.

2.2 Scope and validation

Notebook 01 explicitly excludes the workbook aggregate Total row, the Unknown origin row, and the Canada row classified as Citizens. It standardizes year labels, removes redundant administrative codes, preserves continent, region, and development-group classifications, creates a cumulative Total field, and validates uniqueness, completeness, numeric type, and non-negative annual counts. The resulting cleaned dataset is exported once and reused downstream.

Prepared dataset characteristic	Value
Identified international origins	194
Annual coverage	1980–2013 (34 years)
Country-year observations in long form	6,596
Cumulative identified immigration	6,409,133
Core dimensions	Country, continent, region, development group, year, immigration volume

Table 1. Analysis-ready dataset produced by Notebook 01.

3. Visualization Methodology

The final project is organized into **four notebooks: data preparation; exploratory visual analytics; advanced statistical visualization; and geospatial visualization**. Pandas is used for transformation and aggregation, **Matplotlib** for controlled exploratory figures, **Seaborn** for statistical graphics and heatmaps, and **Folium/GeoJSON** for interactive choropleths.

The report follows the same logic of the notebooks but does not reproduce code. Instead, it groups figures by the analytical question they answer and uses the notebook outputs as empirical evidence.

4. Exploratory Visual Analytics

4.1 Long-run temporal dynamics

The first question is **whether identified international immigration changed systematically over time**. A line chart is appropriate because year is ordered and the analytical focus is trajectory. The series rises substantially over the full window but is not monotonic. Immigration increased from 99,137 in 1980 to 257,535 in 2013, reached a low of 69,978 in **1985**, and peaked at 276,956 in **2010**. The visual therefore supports a phased interpretation rather than a constant-growth narrative.

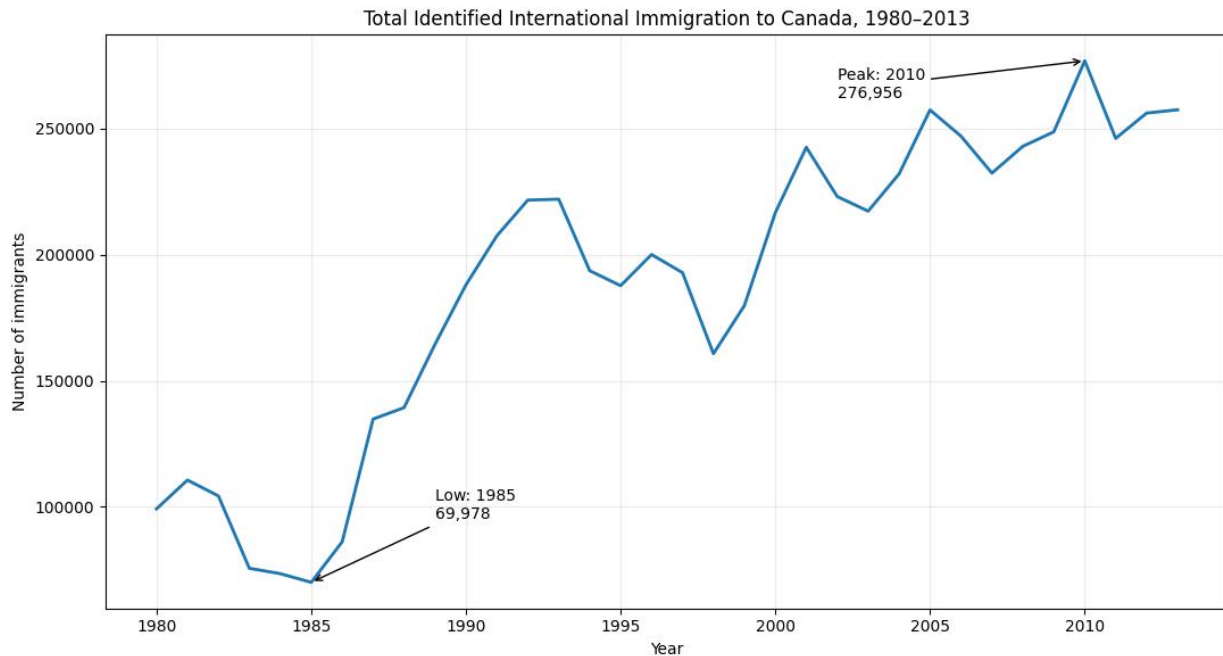


Figure 1. Total identified international immigration to Canada, 1980–2013.

4.2 Source-country hierarchy

Cumulative totals reveal a pronounced source-country hierarchy. **India** (691,904) and **China** (659,962) are the two largest origins, followed by the **United Kingdom** (551,500) and the **Philippines** (511,391). A horizontal bar chart is used because length along a common baseline supports more precise ranking.

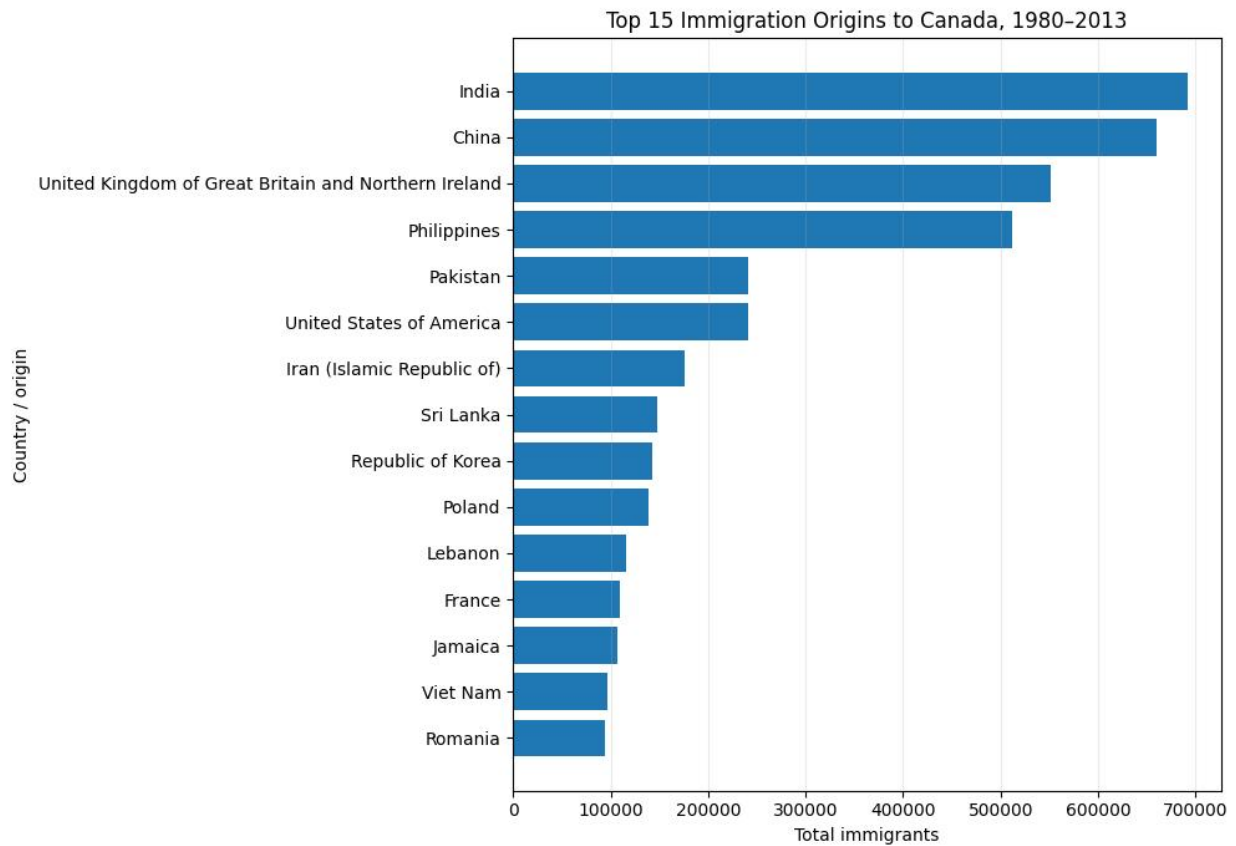


Figure 2. Top 15 immigration origins to Canada by cumulative immigration, 1980–2013.

4.3 Different trajectories among the largest origins

Cumulative rank alone hides timing. The five largest origins followed visibly different annual paths. India and China both became dominant sources but did not peak at the same time; the United Kingdom began from a comparatively high level and declined over much of the period; the Philippines rose strongly later; and Pakistan grew from a much smaller 1980 base. The figure separates cumulative importance from **temporal pattern**.

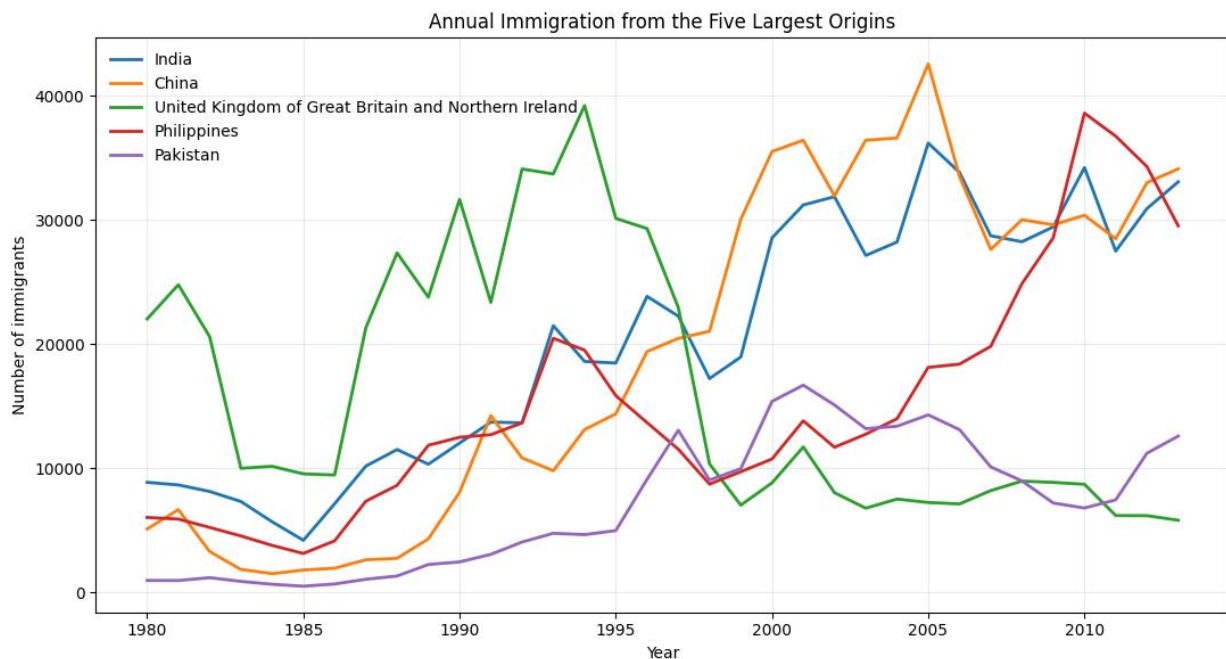


Figure 3. Annual immigration trajectories for the five largest cumulative origins.

4.4 Continental composition through time

Aggregating origins by continent reveals a broader structural change. The stacked area chart simultaneously shows national volume and continental contribution. **Asia** became increasingly dominant, while Europe's relative contribution contracted substantially. This pattern is not visible from the country ranking alone and motivates the normalized statistical views in Section 5.

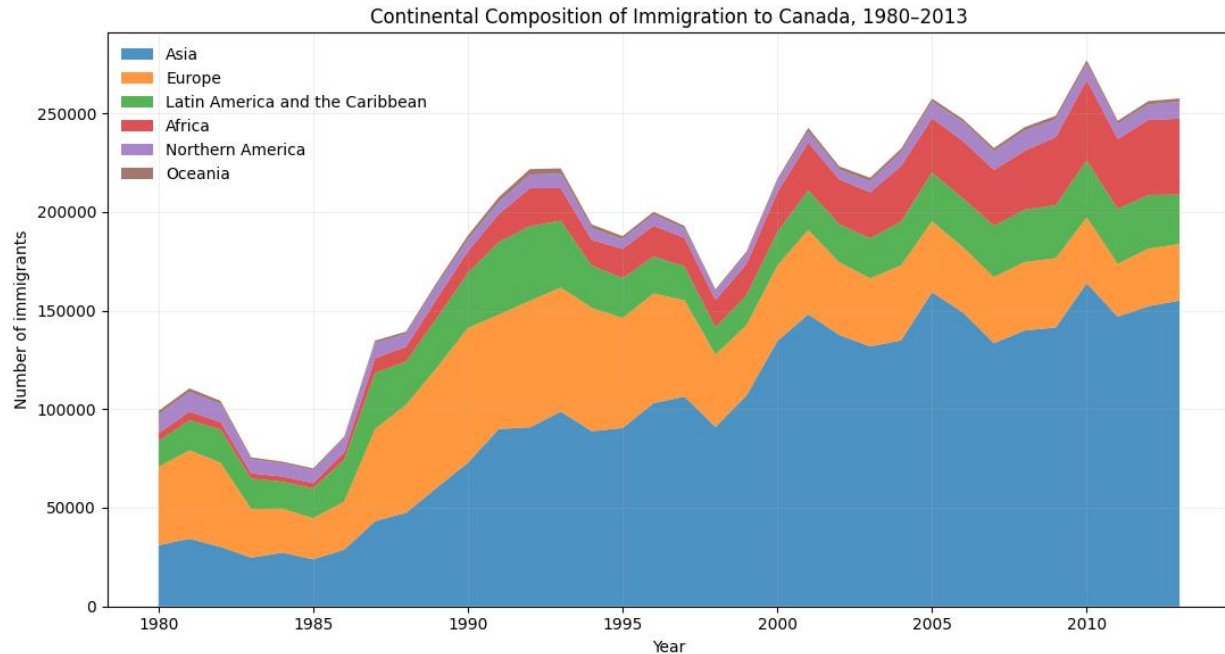


Figure 4. Continental composition of immigration to Canada, 1980–2013.

4.5 Distribution of origin-country counts in 2013

The 2013 country-level distribution is strongly **right-skewed**. The median origin contributed only 215 immigrants, while the maximum was 34,129; 11 origins recorded zero immigrants in 2013. The histogram makes clear that the national total is generated disproportionately by a relatively small upper tail of high-volume origins.

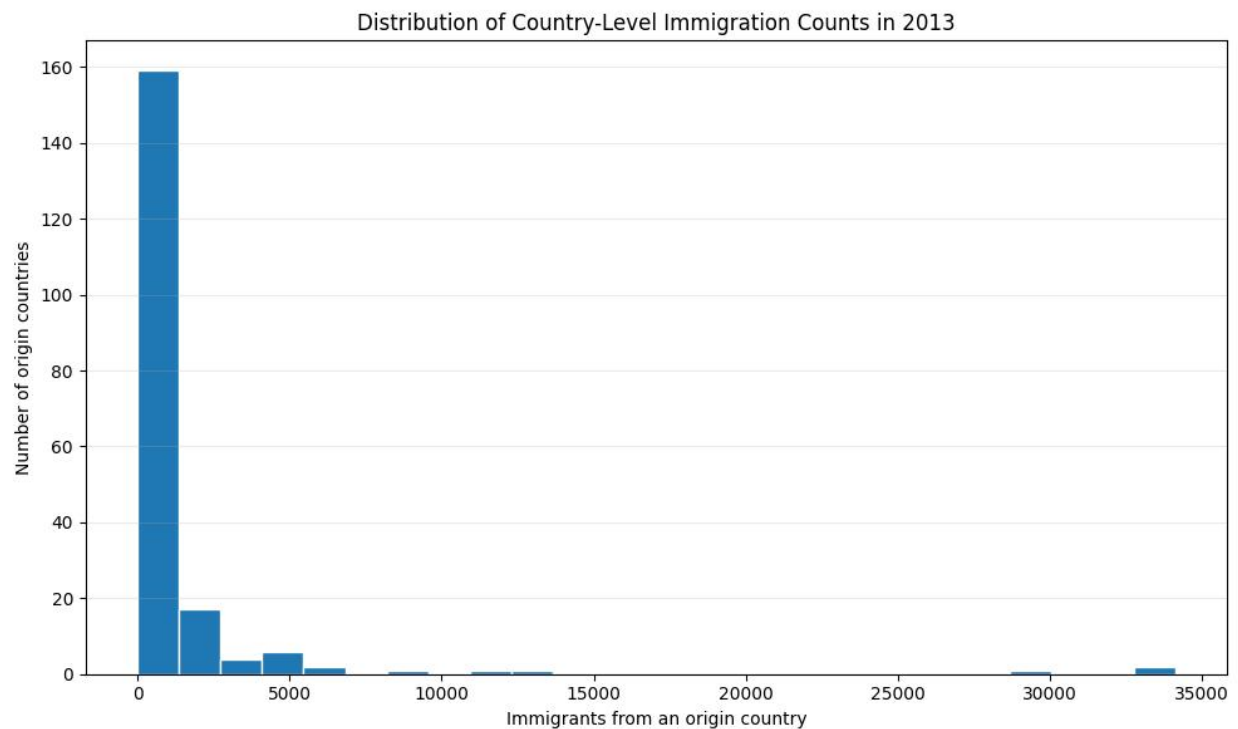


Figure 5. Distribution of country-level immigration counts in 2013.

4.6 Concentration of cumulative immigration

A cumulative concentration curve quantifies the imbalance suggested by the ranking and histogram. Only **eight countries** are required to exceed **50%** of cumulative immigration, while **54 countries** are required to exceed **90%**. This Pareto-style view is analytically useful because it converts visual dominance into an interpretable concentration measure.

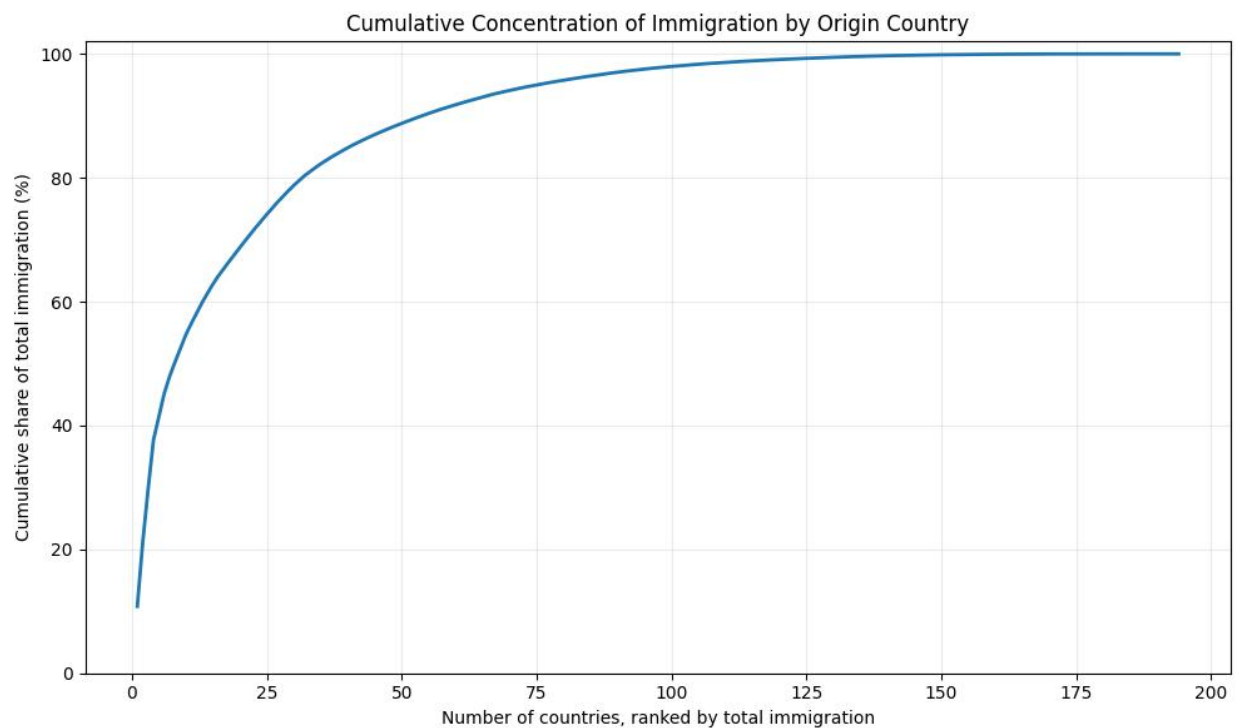


Figure 6. Cumulative concentration of immigration by origin country.

4.7 Scale versus temporal variability

The top 15 origins are compared on mean annual immigration and coefficient of variation, with marker size representing cumulative immigration. This multivariate view demonstrates that scale and stability are separate characteristics: **high-volume origins can be relatively stable or volatile**, and **smaller major origins can show substantial relative variability**. The graphic adds information that neither a ranking nor a time series provides alone.

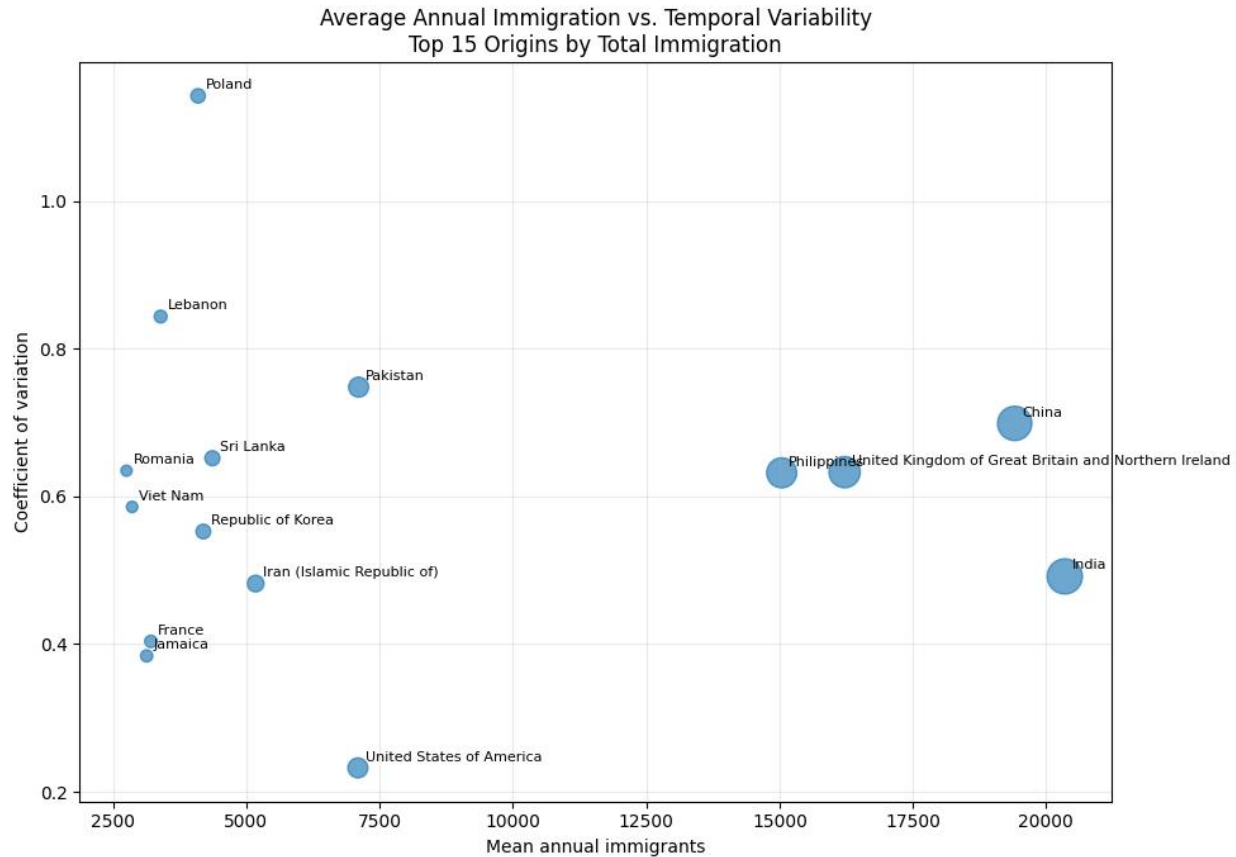


Figure 7. Average annual immigration versus temporal variability for the top 15 origins.

4.8 Event annotation: Haiti

The Haiti case illustrates explanatory annotation. Immigration from Haiti increased from 2,080 in 2009 to 4,744 in 2010 and 6,503 in 2011. The notebook marks the **2010 Haiti earthquake** as contextual information associated with the visible post-2010 rise. The figure is retained to demonstrate how the study can connect a time-series pattern with a historically meaningful event while avoiding a causal claim based on the graph alone.

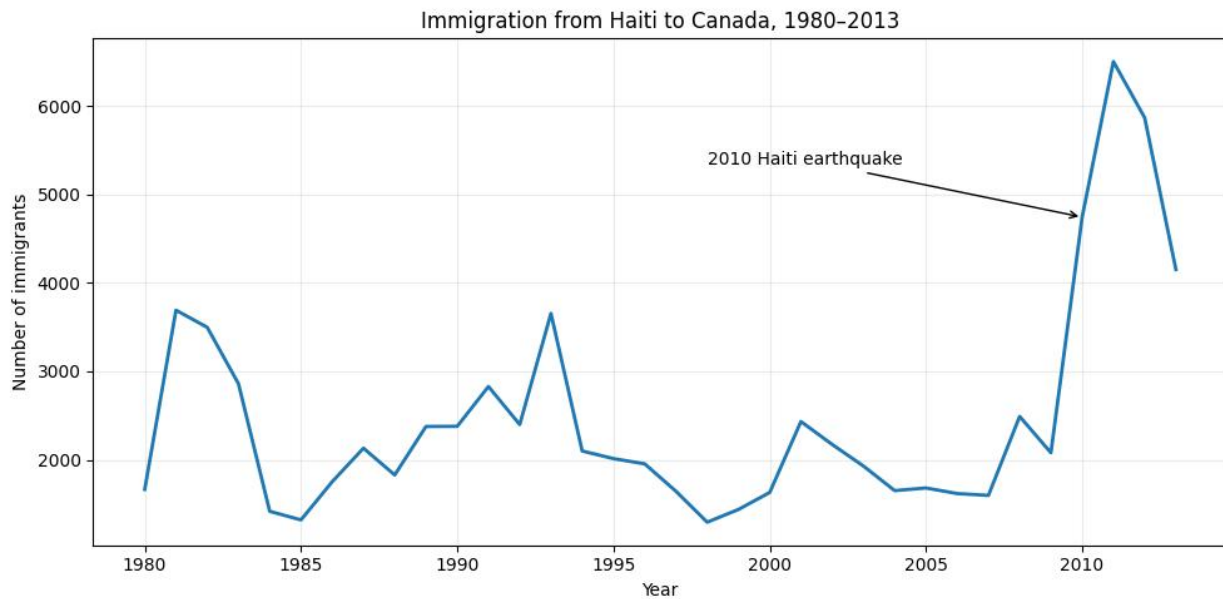


Figure 8. Immigration from Haiti to Canada with annotation for the 2010 earthquake period.

5. Advanced Statistical Visualization

5.1 Distributional differences across continents

The boxplot shifts attention from continent-level totals to the **distribution of country-level cumulative totals within each continent**. A logarithmic scale is used because totals span several orders of magnitude. Asia contains several exceptionally large origins and therefore a strong upper tail; Europe also contains major origins but with a different distributional profile. Northern America contains only one retained origin in this dataset and should not be interpreted as a comparable multi-country distribution.

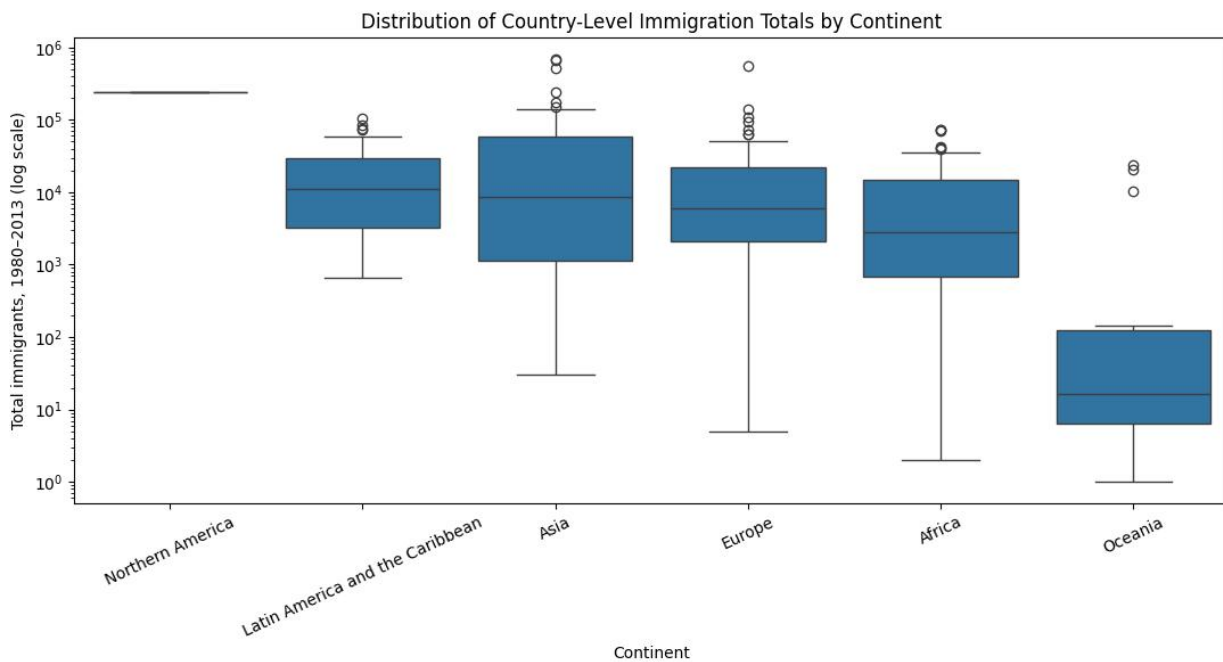


Figure 9. Distribution of country-level immigration totals by continent (log scale).

5.2 Developed versus developing regions

The **violin plot** compares the full distribution of cumulative origin-country totals across **development groups**. The two distributions overlap substantially. Developed regions have a higher median country-level total, but developing regions include several of the largest individual origins. The figure therefore cautions against treating development classification as a simple separator of immigration magnitude.

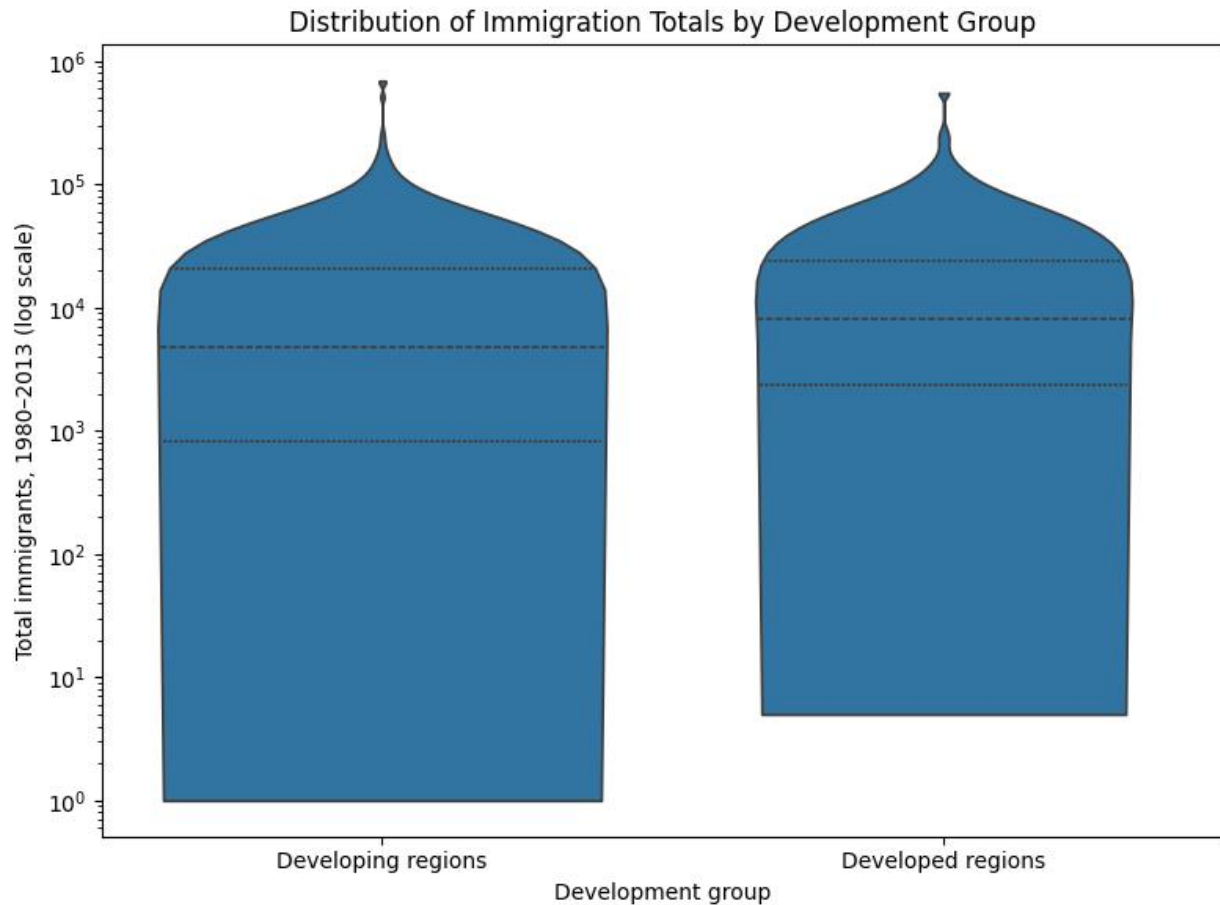


Figure 10. Distribution of immigration totals by development group (violin plot, log scale).

5.3 Normalized continental composition

The percentage **heatmap** normalizes selected years so that each column sums to 100%, making compositional change easier to compare. **Europe** represented approximately **40.1%** of identified international immigration in **1980**, compared with 31.3% for Asia. By **2013**, **Asia's** share had risen to **60.2%**, while Europe's had fallen to 11.1%. Africa's share also increased over the period. The heatmap therefore reveals structural rebalancing rather than simply growth in absolute volume.

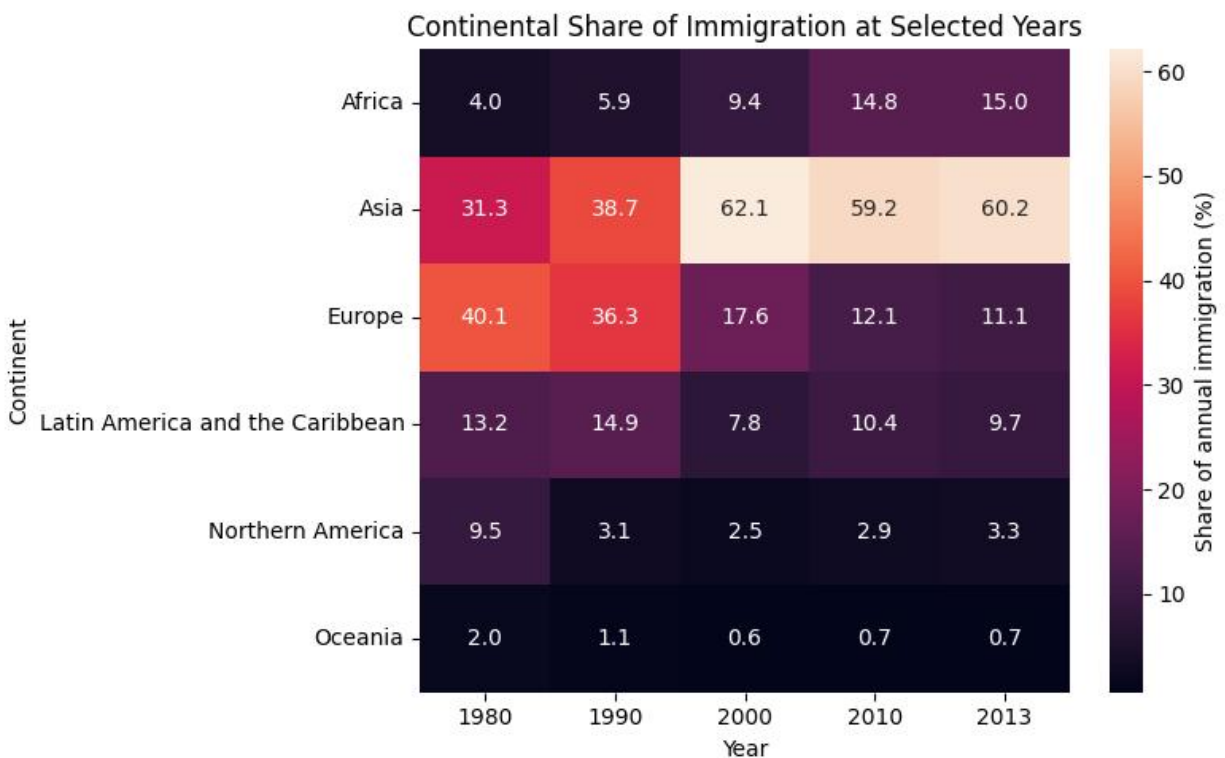


Figure 11. Continental share of immigration at selected years.

5.4 Standardized temporal patterns among major origins

Raw counts make high-volume origins visually dominant. To compare timing rather than scale, each of the top ten origins is standardized within country using **z-scores**. The **heatmap** shows that **major origins reached unusually high relative levels in different periods**—some in the early 1990s, others in the 2000s, and others near the end of the series. This is a high-dimensional view of temporal structure that complements the raw top-five line chart.

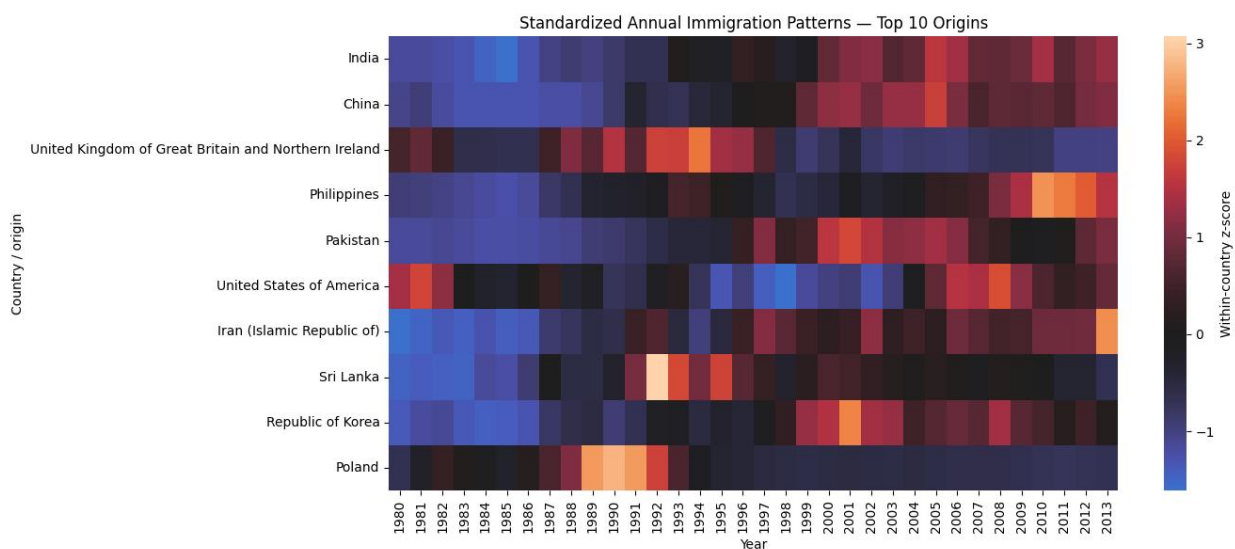


Figure 12. Standardized annual immigration patterns for the top 10 origins.

5.5 Descriptive linear trend

A **Seaborn regression** visualization summarizes the broad association between **year** and **total annual immigration**. The Pearson correlation is **0.897**, indicating a **strong positive association across the 34-year**

period. However, observed points visibly depart from the fitted line, reinforcing that the series contains peaks, reversals, and phases. The line is therefore treated as a descriptive visual summary rather than a forecasting or causal model.

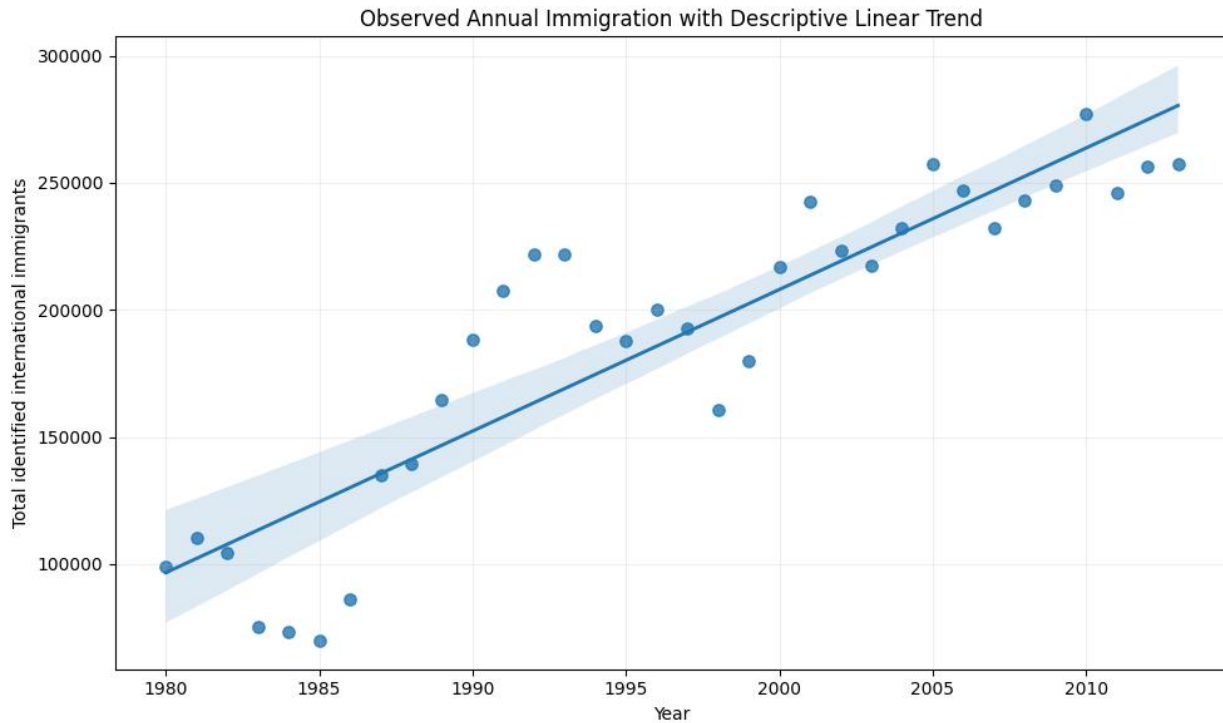


Figure 13. Observed annual immigration with descriptive linear trend.

5.6 Continental co-movement

The **correlation heatmap** compresses pairwise relationships among annual continental immigration totals into one matrix. **Positive correlations indicate broad co-movement over time**; weaker or negative relationships indicate different long-run paths. Because all series share the same national immigration environment and contain time trends, the matrix is interpreted as exploratory association only, not evidence of causal dependence.

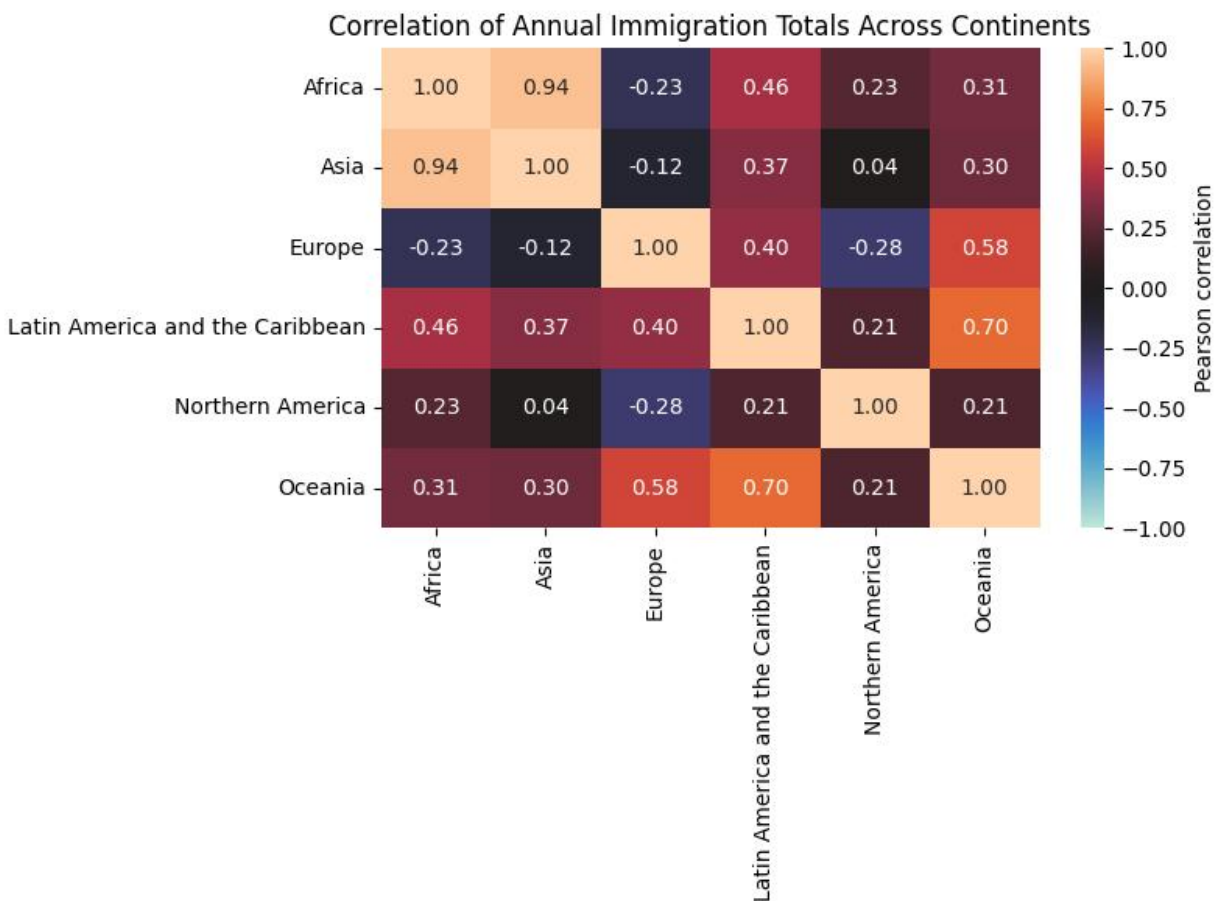


Figure 14. Correlation of annual immigration totals across continents.

6. Geospatial Visualization

6.1 Geographic linkage and map coverage

The immigration data are linked to a **Natural Earth low-resolution world boundary layer** after transparent country-name harmonization. Of 194 origin records, 164 have independent polygons in the boundary file, yielding record coverage of 84.5%. More importantly for substantive interpretation, mapped origins represent 98.77% of cumulative immigration volume. Unmapped cases are primarily small states, islands, or special territories, so the maps capture nearly all volume while explicitly documenting incomplete record coverage.

6.2 Cumulative global geography, 1980–2013

The **cumulative choropleth** adds spatial structure to the country ranking. High-volume origins are geographically clustered, especially across **Asia**, while substantial contributions are also visible in parts of **Europe**, **North America**, and selected countries elsewhere. Geographic adjacency and regional clustering are immediately perceptible in a way that a ranked table cannot provide.

Total Immigration to Canada by Country of Origin, 1980–2013

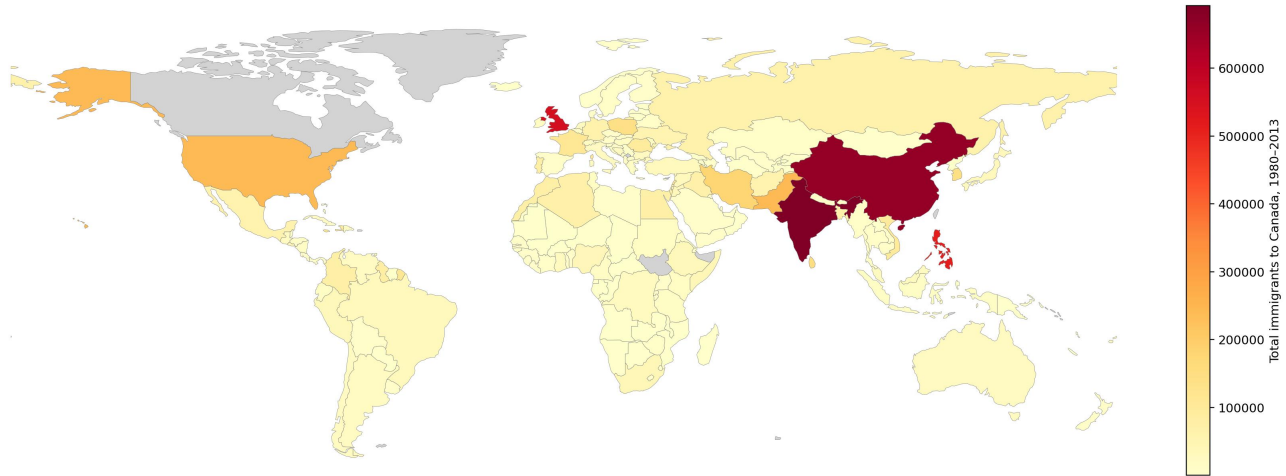


Figure 15. Total immigration to Canada by country of origin, 1980–2013.

6.3 Geographic pattern in 2013

A separate **2013 choropleth** prevents the 34-year cumulative map from obscuring later-period structure. The end-of-series geography makes the increased importance of **Asian** source countries particularly visible and demonstrates that the dominant origins in a historical cumulative total need not be identical to the strongest origins in a single later year.

Immigration to Canada by Country of Origin, 2013

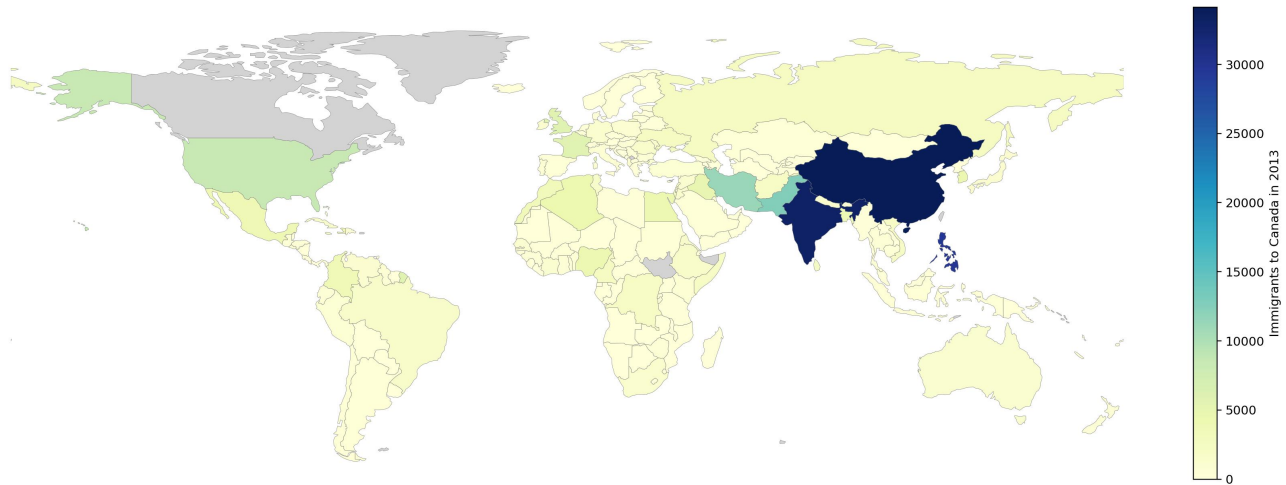


Figure 16. Immigration to Canada by country of origin in 2013.

6.4 Change in immigration share, 1980–2013

The final map focuses on relative rather than absolute change. Each country's **share** of all identified international immigration in **2013** is compared with its **share in 1980**. **Positive values indicate increased importance in Canada's immigration mix; negative values indicate declining share**. This representation distinguishes general growth in national immigration from structural reallocation among source countries.

The largest gains include the **Philippines (+5.37 percentage points)**, Pakistan (+3.91), India (+3.89), and Iran (+3.20). The largest decline is the **United Kingdom (-19.97 percentage points)**, followed by the United States (-6.16) and Portugal (-4.27).

Change in Share of Canadian Immigration by Country, 1980–2013

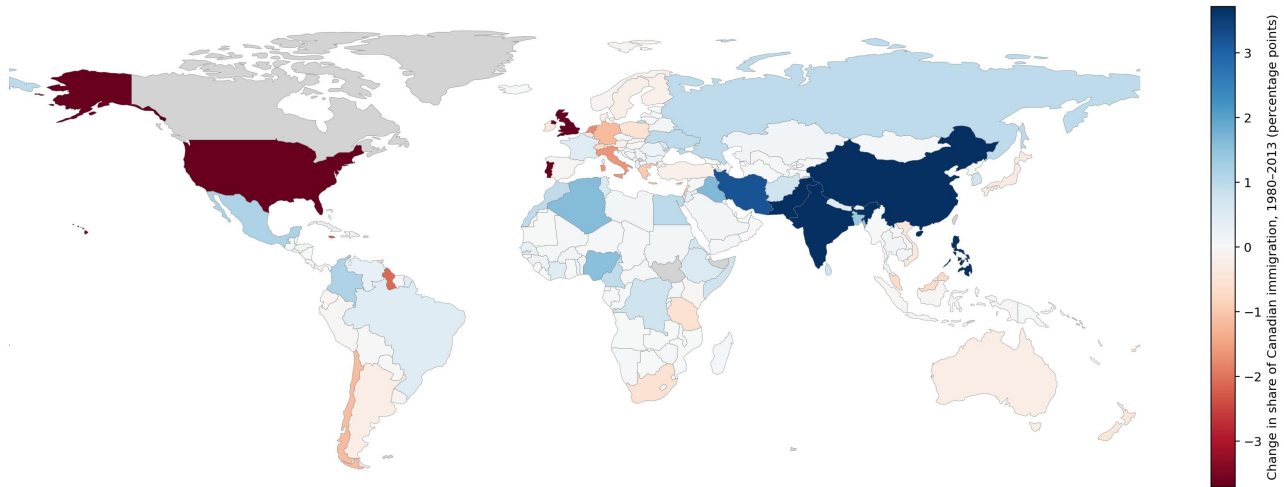


Figure 17. Change in share of Canadian immigration by country, 1980–2013.

7. Integrated Empirical Findings

Viewed together, the 17 figures provide a layered account of **Canadian immigration flows**. The line chart establishes long-run growth but also temporal irregularity. Ranked and concentration graphics show that the total is disproportionately driven by a small group of origins. Country trajectories and standardized heatmaps show that major origins do not share one common timing pattern. Distributional graphics reveal strong right-skew and within-group heterogeneity. Continental area and percentage heatmaps demonstrate a major compositional shift toward Asia. Finally, choropleths show that the statistical rebalancing is also a geographic rebalancing.

- **Long-run growth was substantial but non-linear:** 99,137 immigrants in 1980, 257,535 in 2013, and a peak of 276,956 in 2010.
- **India, China, the United Kingdom, and the Philippines** were the four largest cumulative origins.
- **Concentration was high:** eight origins exceeded 50% of cumulative immigration and 54 exceeded 90%.
- The 2013 origin distribution was highly **right-skewed**, with a median of 215 and maximum of 34,129.
- **Asia's** annual share increased from 31.3% in 1980 to 60.2% in 2013; **Europe's** fell from 40.1% to 11.1%.
- A **simple linear trend** captures broad growth ($r = 0.897$ with year) but not the observed phases and reversals.
- **Geospatial** linkage represented 98.77% of cumulative immigration volume, supporting robust world-level spatial interpretation despite missing small territories.

8. Relevance to Business and Data Analytics

The project's value for a business/data analytics portfolio lies in the **analytical workflow** rather than the immigration domain alone. It demonstrates the ability to convert a raw workbook into a validated analytical asset, select visual encodings based on decision questions, compare segments and distributions, quantify concentration, distinguish scale from volatility, normalize data for fair comparison, integrate geographic data, and communicate limitations alongside findings.

- **Trend monitoring:** analogous to revenue, demand, traffic, or KPI analysis over time.
- **Ranking and concentration:** analogous to customer, supplier, product, or market concentration analysis.
- **Distributional comparison:** analogous to segment heterogeneity, risk dispersion, or operational variability.
- **Normalization and heatmaps:** useful for comparing units with different scales and identifying structural patterns.

- **Geospatial analytics:** transferable to market expansion, site selection, regional performance, logistics, and customer geography.
- **Communication discipline:** the notebooks preserve reproducibility while the report converts technical output into an interpretable research narrative.

9. Limitations and Analytical Boundaries

The analysis is **descriptive** and does not establish causal mechanisms. Immigration flows may reflect policy, economic conditions, geopolitical events, labor-market demand, family reunification, refugee movements, demographic change, and data-collection practices that are not modeled here.

- The source reports immigration flows, not immigrant stock or resident population at a point in time.
- The project uses citizenship/origin classifications; these should not be conflated with ethnicity or place of birth.
- The dataset ends in 2013, so the findings are historical rather than claims about current Canadian immigration.
- International migration-flow statistics can face comparability issues because national statistical systems use different definitions and concepts.
- The Natural Earth low-resolution layer omits some small states and territories; record-level mapping coverage is 84.5%, although volume coverage is 98.77%.
- Choropleths visually privilege large geographic polygons, so maps should be interpreted with ranked and statistical figures.
- Regression and correlation figures are exploratory summaries; trending time series can produce strong associations without causal dependence.

10. Reproducibility and Portfolio Architecture

Component	Primary role	Key output
01 Data Preparation	Raw Excel ingestion, scope definition, validation, export	Cleaned analysis-ready dataset
02 Exploratory Visual Analytics	Trends, rankings, distributions, concentration, variability	Eight question-driven Matplotlib figures
03 Advanced Statistical Visualization	Distributional comparison, normalization, association	Six Seaborn statistical graphics / heatmaps
04 Geospatial Visualization	Boundary linkage, choropleths, share change	Three interactive Folium maps plus HTML exports

Table 2. Final project architecture.

The three **Folium maps** are exported as standalone **HTML files** for interactive use. The PNG maps embedded in this report are publication-oriented static views; the HTML versions preserve hover tooltips, zooming, panning, legends, and layer interaction.

11. Conclusion

This project demonstrates **visualization as an analytical method**. Starting from a raw UN workbook, the workflow creates a validated data foundation, uses exploratory graphics to identify trend and concentration, applies statistical visualization to reveal distribution and normalized temporal structure, and concludes with interactive geospatial analysis of global origin patterns.

The rich figure set is important to the portfolio's purpose. No single chart establishes the full empirical story: time-series figures reveal phases, rankings reveal dominance, histograms and box/violin plots reveal heterogeneity, concentration curves quantify imbalance, heatmaps reveal composition and standardized timing, and choropleths reveal spatial structure. Together, these complementary views demonstrate both technical breadth in **Python visualization** and **judgment about when each visual representation is analytically appropriate**.

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Natural Earth. Natural Earth vector map data (low-resolution world boundaries). <https://www.naturalearthdata.com/>

Appendix A. Selected Exact Computational Results

Metric	Executed result
1980 annual immigration	99,137
2013 annual immigration	257,535
Peak annual immigration	276,956 (2010)
Minimum annual immigration	69,978 (1985)
Largest cumulative origin	India: 691,904
Second-largest cumulative origin	China: 659,962
Median origin count in 2013	215
Maximum origin count in 2013	34,129
Origins with zero recorded immigration in 2013	11
Countries required to exceed 50% cumulative share	8
Countries required to exceed 90% cumulative share	54
Pearson correlation: year vs annual immigration	0.897
Mapped origin records	164 / 194 (84.5%)
Mapped cumulative immigration volume	98.77%